

BENCHMARK: ACCURATE — fit consistent with published values

Results

|                                                   |                                  |
|---------------------------------------------------|----------------------------------|
| Mid-Transit Time (Tmid)                           | 2458508.8542 +/- 0.00049 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.111 +/- 0.0063                 |
| Transit depth (Rp/Rs)^2                           | 1.23 +/- 0.14 [%]                |
| Orbital Inclination (inc)                         | 86.1 +/- 0.3                     |
| Airmass coefficient 1 (a1)                        | 1.2355 +/- 0.0064                |
| Airmass coefficient 2 (a2)                        | -0.1198 +/- 0.0044               |
| Scatter in the residuals of the lightcurve fit is | 0.52 %                           |
| Best Comparison Star                              | None                             |
| Optimal Aperture                                  | 5.22                             |
| Optimal Annulus                                   | 14.3                             |
| Transit Duration (day)                            | 0.1222 +/- 0.0043                |

Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                          |
|---------------------------|------------------------------------------|
| Rp/R■ fit vs published    | 0.111 ± 0.0063 vs 0.11 (deviation 0.09σ) |
| Duration fit vs published | 0.1222 d vs 0.1208 d (deviation 0.17σ)   |
| Mid-time O–C              | -1.3 min                                 |
| Depth SNR                 | 9.6                                      |
| Residual scatter          | 0.52 %                                   |

Light curve

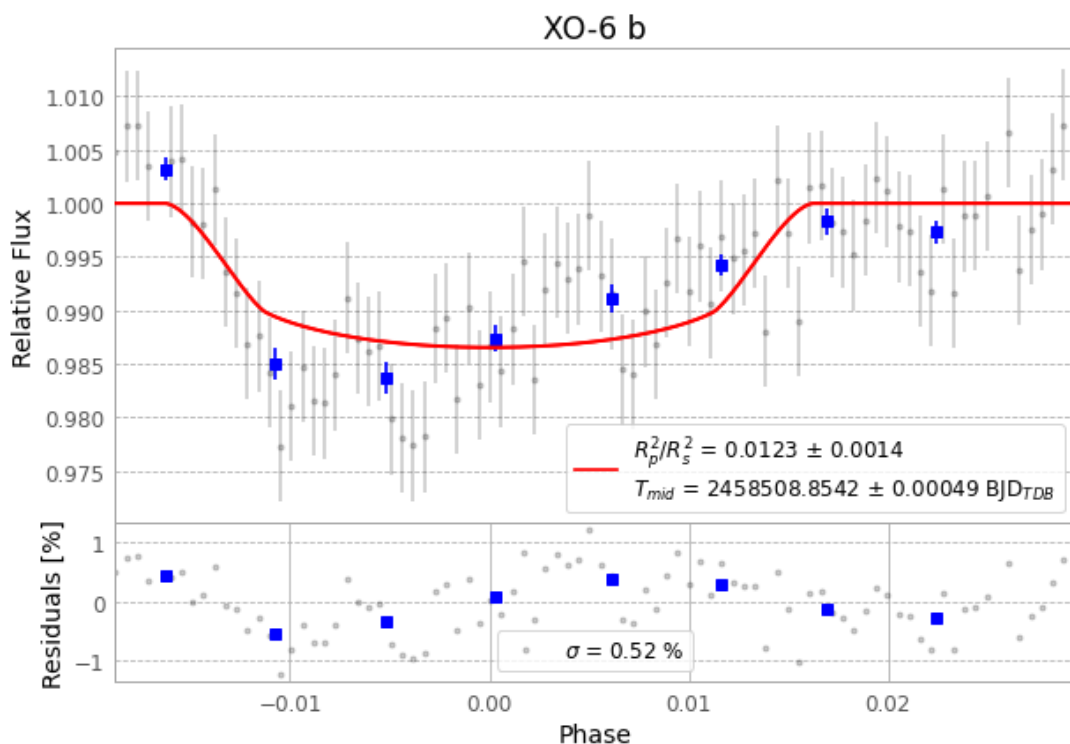


Figure 1 — FinalLightCurve\_XO-6 b\_2019-01-24.png (SHA-256 1e24d4c8748a96f7...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [498, 350], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 555b1d0c93684749...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit 1dc3d22

## Data provenance

88 FITS frames (58 MB), XO-6190125064212.FITS → XO-6190125110313.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-XO-6-190125.log (8708b38ba782...), FinalLightCurve\_XO-6 b\_2019-01-24.png (1e24d4c8748a...), FinalLightCurve\_XO-6 b\_2019-01-24.csv (4a5a684b8d2c...)

## Compute resources

Wall time 205.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

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Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-16 23:55Z.